Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “system” to explain your answer. Sentence starters are optional.

component: the part of a vector along one chosen axis

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

Two carts approach: one has momentum **5 kg·m/s** *east* and the other **3 kg·m/s** *west*. **Explain why adding their magnitudes** gives the wrong total; **find the signed total**.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: ① ② ③

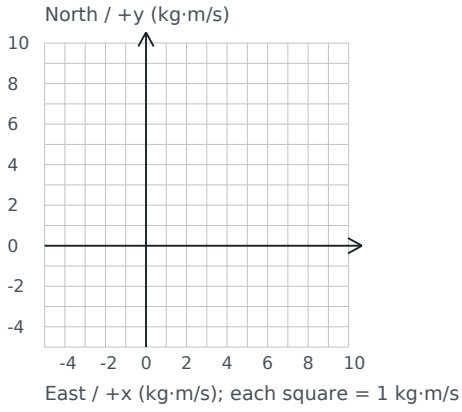
02 Worked example: explain the prediction

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$. Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (**3**, **0**) and (**0**, **4**) kg·m/s. Their total is (**3**, **4**), magnitude **5 kg·m/s**. **Add components before finding the** magnitude.

Your turn: Draw a tip-to -tail momentum diagram and **label each axis** with units.

Sentence starter: In my work, I used … to show … . I labeled it … .



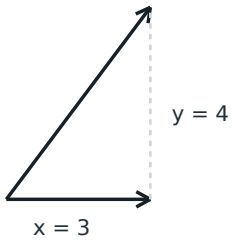
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two incoming arrows point *east* and *north*.
1. **Sketch the direction of their** vector sum.
2. Can their ordinary lengths be added for the diagonal magnitude?

Word help: magnitude = size of a quantity without its direction

Sentence starter: In my work, I used … to show … . I labeled it … .



The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: ① ② ③

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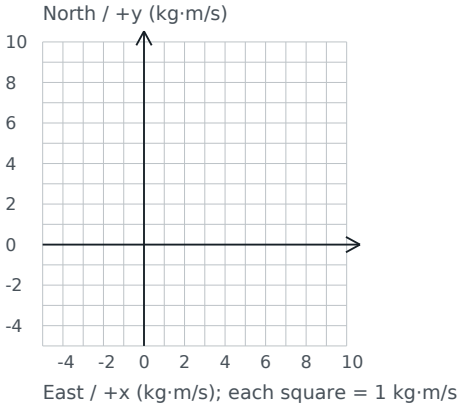
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

Incoming total momentum is **6 kg·m/s** *east* and **8 kg·m/s** *north*. Two objects stick with total mass **2 kg**. Divide each component by **2** to **find velocity components**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

Why must both momentum component sums use the same system and same time interval?

Sentence starter: I think … . I can point to …, which shows …, so … .

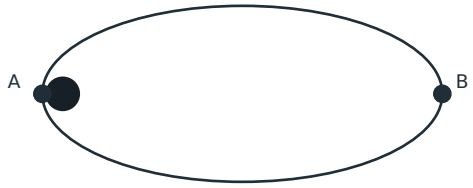
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An ideal asteroid orbit has nearest distance **4 AU** and farthest distance **8 AU**. Where is it fastest? Explain with the equal-area law.
Label the orbital features used in your explanation.

Word help: orbit = a path around another body under gravity

Sentence starter: My design includes … . It does what is needed by … .

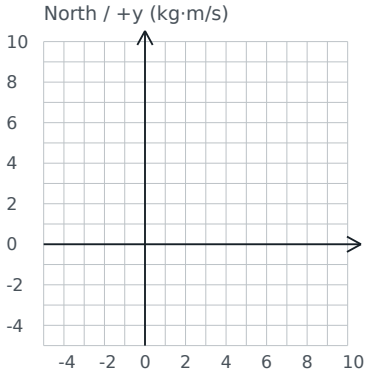


Marzano check - circle: ① ② ③

07 Exit check

Total momentum is **4 kg·m/s** *west* and **2 kg·m/s** *south*. Combined mass **2 kg**. **Give final velocity components**.

Sentence starter: I notice … . This supports … because … .

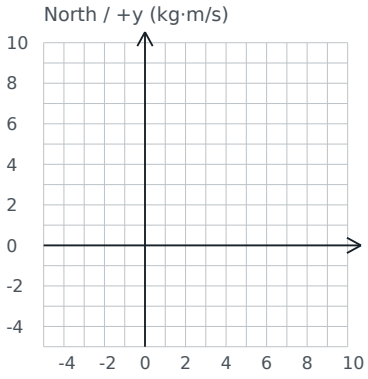


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A joined pair of total mass **2 kg** has total momentum **6 kg·m/s** *east* and **8 kg·m/s** *north*. **Find its velocity components** using each momentum component divided by **2**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③